

Weiming ZHAO

Chief AI Architect / AI for Science

+86 158-0925-5970

✉ weiming.zhao@sjtu.edu.cn

📄 <https://zwm0426.github.io>

Professional Experience

- 2026.06 – Present **Chief AI Architect**, *EC-ZERO Lab, Global Institute of Future Technology*, Shanghai Jiao Tong University, Shanghai, China.
- Lead AI4Science architecture for electrochemistry and energy research, developing AI-assisted research tools and workflows.
 - Build data, model, and knowledge infrastructure for collaboration, experiment management, and laboratory operations.
 - Lead LabOS design and development for experimental workflows, asset traceability, and AI-enabled laboratory infrastructure.
- 2019.06 – **Director of XR Lab**, *Student Innovation Center*, Shanghai Jiao Tong University, Shanghai, China.
- 2026.04
- Led and contributed to university-wide courses in VR/AR, data science, AI, and engineering practice, covering Unity, Python, computer vision, Transformers, and LLM fine-tuning.
 - Led university-industry education projects with Huawei and ByteDance, including four Ministry of Education collaborative-education projects.
 - Managed the XR Lab, including equipment planning, procurement, safety, BodyCapture4D, and a 6-DoF driving simulator; maintained 100% equipment availability and zero safety incidents.
 - Advised the SJTU AIGC Class and micro-program; supervised 50+ student teams across courses, innovation projects, and national competitions, with multiple award-winning teams.
- 2017.11 – **Research Associate**, *Energy Research Institute @ NTU (ERI@N)*, Singapore.
- 2018.06
- Contributed to a large-scale 3D city-visualization project, processing LiDAR point clouds and visualizing urban roads and buildings.
 - Designed an energy-data visualization system using .NET and Google Maps API, integrating sensor data, geospatial information, and interactive dashboards.

Education

- 2016 – 2017 **M.Sc. in Digital Media Technology**, *Nanyang Technological University*, Singapore, GPA: 4.19 / 5.00.
- 2012 – 2016 **B.Sc. in Computer Science and Technology**, *Northwest University*, Xi'an, China, GPA: 3.35 / 4.00.

Selected Publications

- [1] Xuanchen Li, Jianyu Wang, Yuhao Cheng, Yikun Zeng, Xingyu Ren, Wenhan Zhu, **Weiming Zhao**, Yichao Yan. "Towards High-fidelity 3D Talking Avatar with Personalized Dynamic Texture." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [2] Zhuo Chen, Yichao Yan, Shengqi Liu, Yuhao Cheng, **Weiming Zhao**, Lincheng Li. "Revealing Directions for Text-guided 3D Face Editing." *IEEE Transactions on Multimedia*, 2025. DOI: 10.1109/TMM.2025.3604978.

- [3] Weili Zeng, Yichao Yan, Qi Zhu, Zhuo Chen, Pengzhi Chu, **Weiming Zhao**, Xiaokang Yang. "Infusion: Preventing Customized Text-to-Image Diffusion from Overfitting." *ACM International Conference on Multimedia (ACM MM)*, 2024.
- [4] Yinfan Liu, Yichao Yan, Junjie Li, **Weiming Zhao**, Pengzhi Chu, Xingdong Sheng, Yunhui Liu, Xiaokang Yang. "IPAD: Industrial Process Anomaly Detection Dataset." *IEEE Transactions on Circuits and Systems for Video Technology*, 2024.
- [5] Calibration Method of Three-Dimensional Point Cloud Acquisition System. Chinese Patent No. CN110196031B. Inventors: Lin Wang, Zewei Ding, Wanxu Zhang, **Weiming Zhao**, et al. Assignee: Northwest University, China.
- [6] Pengzhi Chu, Bo Xu, Xiongziyan Xiao, **Weiming Zhao**, Wankun Xue. "The Design of Autonomous Driving Experimental Platform Based on the Huawei Cloud ModelArts and HiLens." *Drive System Technique*, 36(2), 38–39, 2022.

Selected Honors & Awards

- 2025 Huawei Cloud University–Industry Collaboration Teaching Excellence.
- 2025 Selected for the Shanghai Jiao Tong University "Excellence Program" (Laboratory Engineer track).
- 2025 Grand Prize, 14th Shanghai Undergraduate Engineering Practice and Innovation Competition.
- 2024 Outstanding Advisor, China International College Students' Innovation Competition (Shanghai Division).
- 2020 Excellent Staff in Annual Assessment, Shanghai Jiao Tong University.
- 2020 Grand Prize, Second National University Blended Learning Design and Innovation Competition.

Selected Teaching

- CS1107 **VR/AR Design and Development Practice**
Course Lead (2021–2025); rating 5.0/5.0; top 5% at SJTU.
- CS1108 **Introduction to Data Science: The Way of Python**
Co-instructor / Course Lead (2023–2025); rating 4.5/5.0; top 15% at SJTU.
- AI1113 **Artificial Intelligence: Foundations and Practice**
Module Lead (2025); rated in top 10% at SJTU; LLM fine-tuning practice.

Technical Expertise

- AI & Scientific Computing Python, PyTorch, LLM workflows and local deployment, LoRA fine-tuning, computer vision, multimodal data processing.
- Interactive Systems Unity, XR/AR prototyping, spatial computing, human–computer interaction.
- Engineering Systems MATLAB, NI LabVIEW, Raspberry Pi, Linux, LiDAR and 3D visualization.
- Languages Mandarin Chinese (native); English (IELTS 7.0).